



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Information Technology

Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: I Semester B.Tech IT

Course Code & Title: GE3151 Problem solving and python programming

Name of the Faculty member (s): Mrs. G. Mareeswari

Innovative Practice Description

Unit / Topic: Unit II / Operators

Course Outcome: CO 2

Topic Learning Outcome: TLO 5

Activity Chosen: Virtual Laboratory

Date of Implementation: 19.10.2023

Justification:

- The Virtual Laboratory is an interactive simulation environment where users may design and carry out simulated experiments. An on-screen calculator or simulator that allows students to test theories and see outcomes is called a virtual laboratory.
- They give the student the opportunity to "tinker" with lab equipment that functions essentially the same as it would in the real world. Students conduct a number of experiments that produce real-world outcomes.
- **Time Allotted for the Activity:** 15 minutes

Details of the Implementation:

- Instructor explained the particular concepts/topic in classroom within 35 minutes. Instructor gave a program to the students. The following link is used to access the virtual lab.
- <https://python-iitk.vlabs.ac.in/> URL for Virtual lab
- The students were instructed to choose Computer Science and Engineering tab that provide link to the page contains broad area of virtual labs
- Select Python Programming Lab from the listed subject
- Choose List of Experiments from the drop down list.
- Select arithmetic operations and the students were asked to answer the pretest.
- Procedure for steps for simulator is explained clearly.
- Demonstrate the operations using simulation button

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO 2	2	1	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO5	PO9	PSO1
	2	1	1	1	1	1
Justification for correlation	Students can able to apply logical thinking and use basic mathematical skill to solve the problem.	Students will be able to use this operator concept to solve simple and complex engineering problems.	Students will be able to interpret the complex engineering problems using the precedence of operators.	Student use this modern tool Virtual lab to understand various topics in detail. Vlab covers lot of topics in problem solving and python programming.	Here students can able to actively involve as an individual and got clear understanding about operator	The problem solving skill earned through this activity helps the students in solving various computational problems.

- **Images / Screenshot of the practice:**

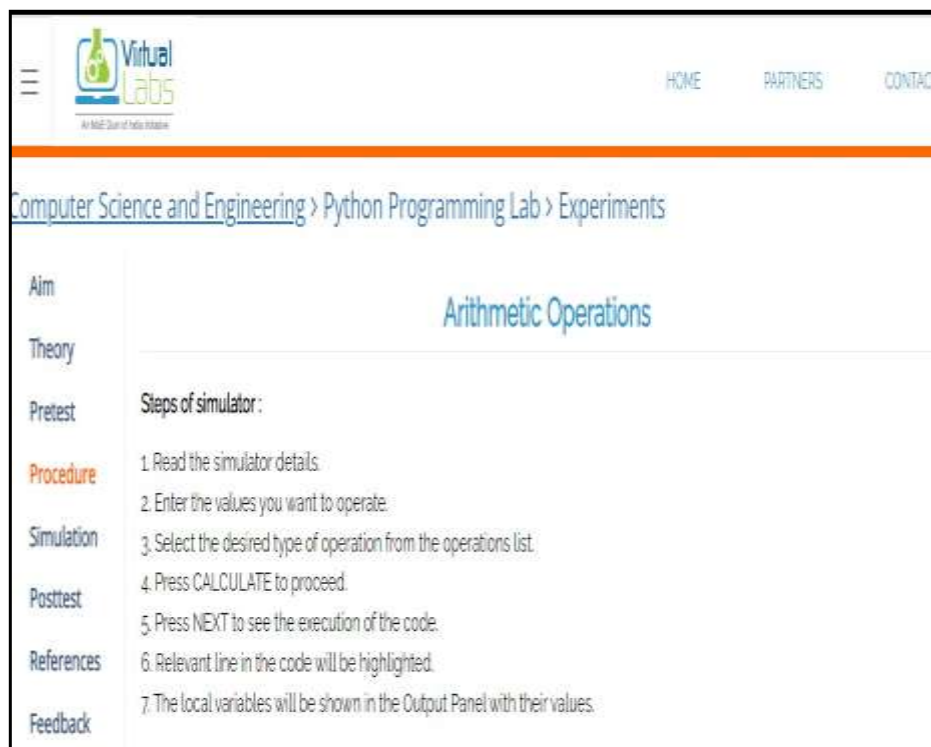


Fig 1: Demonstration on operators

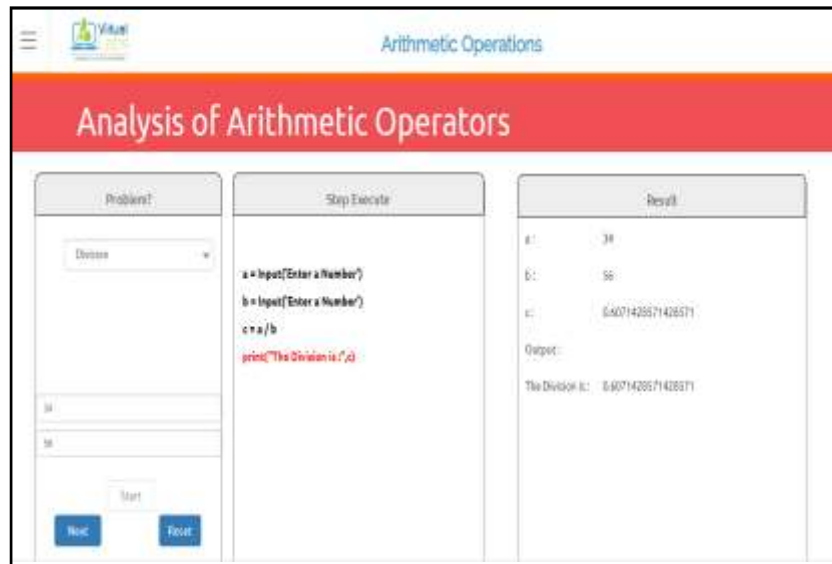


Fig:2 Demonstration on analysis of arithmetic operators



Fig: 3 Mr. Dinesh Kumar S attend post quiz of virtual lab



Fig:4 Students recollect the concept of operator and usage of virtual lab

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- This helped the students to understand the basics of Arithmetic Operations used in Python Programming Language with the help of an iterative simulator.
 - Students said that the pretest and posttest is very useful to recollect the concepts.

- ❖ ***Benefit of the practice:***

- Based on the activity student's felt that it was easy to understand the concept.
 - Most of the students were actively participated and enjoyed once they answered correctly in pretest.

- ❖ ***Challenges faced in implementation:***

- Some students only involved in the demonstration activity.
 - The same activity will be planned in the laboratory session to make all the students familiar in the virtual environmaent.

References:

1. <https://python-iitk.vlabs.ac.in/exp/arithmetic-operations/index.html>
2. <https://www.ritrjpm.ac.in/images/computer-science/Virtual%20Lab-Routing%20in%20MANET.pdf>
3. https://www.ritrjpm.ac.in/images/computer-science/40.EC8393_virtual%20lab.pdf

Signature of Faculty Member

HOD